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ANNOTATED BIBLIOGRAPHY

Is the Best, the Best? Analysing Best-Paper Awardees' Career Trajectories

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Contents

Introduction

This annotated bibliography covers the literature underpinning the thesis. It is organised into five areas:

1. Awards and post-award productivity
2. Cumulative advantage
3. Early-career predictors
4. Disruption and collaboration
5. Data and methods

1 Awards & Post-Award Productivity

Impact Of Major Awards On The Subsequent Work Of Their Recipients

Nepomuceno et al. examine how major scientific prizes, including the Nobel Prize, MacArthur Fellowships, and similar high-prestige awards affect the subsequent publication output and citation impact of recipients. Analysing large-scale bibliometric records, they find that awardees frequently experience a decline in citations per publication after receiving recognition, suggesting that the nature or direction of their work shifts following the award event [1].

This paper is the most directly relevant starting point for the thesis because it establishes the empirical pattern, post-award productivity shifts that motivates the entire project. However, as the authors focus exclusively on senior scientists receiving rare global prizes, it raises the central question my thesis addresses, whether the same dynamics hold for junior researchers receiving frequent, conference-level awards in computer science.

Prizes And Productivity: How Winning The Fields Medal Affects Scientific Output

Borjas and Doran study Fields Medal recipients, the highest award in mathematics, and compare their post-award research output to a carefully constructed counterfactual group of similar mathematicians who did not receive the medal. They found that Fields Medalists reduce their publication rate after winning and shift their research toward newer, riskier topics, suggesting that the medal changes both the quantity and nature of scientific contributions [2].

The matching design employed here, omparing award recipients against similar non-recipients, directly informs the methodological approach of my thesis. Their use of a control group to isolate post-award effects from pre-existing trends is the blueprint for the matched cohort construction planned in this project. The finding that awards may alter research direction rather than simply rewarding it is a key theoretical anchor.

Earlier Recognition Of Scientific Excellence Enhances Future Achievements And Promotes Persistence

Zhu et al. investigate the effects of early-career recognition on scientific output, comparing scientists who received prizes relatively early in their careers to matched peers who did not. Contrary to findings on senior prize winners, they report that early awardees publish significantly more, receive more citations, and remain active in their fields longer than comparable non-recipients, providing strong evidence for a cumulative advantage mechanism [3].

This paper forms a critical counterpoint to and within the thesis's literature review, recognition at an early career stage may operate by different mechanisms than recognition at a late one. The matched comparison design and focus on early-career stage closely mirror my thesis's own approach, and the positive outcome findings motivate testing whether conference best-paper awards, an even more frequent and field-specific form of early recognition show similar patterns.

Superstar Extinction

Azoulay et al. study the effects of the sudden death of eminent "superstar" life scientists on their close collaborators' subsequent productivity. Using a natural experiment framework, they find that collaborators experience a 5–10% permanent decline in publication output following the loss of a superstar, with the effect concentrated among those whose research areas overlapped most with the deceased, indicating that knowledge and intellectual mentorship not just network ties drives scientific productivity [4].

Although this paper studies productivity loss rather than recognition gain, it is highly relevant because it demonstrates the causal role that collaboration relationships play in shaping scientific output. For my thesis, it motivates examining co-authorship network evolution for junior awardees, if gaining access to prominent collaborators (via award visibility) can boost output, this mechanism may help explain any post-award productivity advantages.

Does Science Advance One Funeral At A Time?

This paper extends the "superstar extinction" line of inquiry by asking whether the death of an eminent scientist ultimately benefits a field by creating space for new entrants. Azoulay et al. find that while collaborators' output falls, the broader field eventually benefits as previously marginalised researchers, those outside the superstar's network begin to publish more, suggesting that elite dominance can inhibit outsiders [5].

For my thesis, the paper provides evidence that scientific recognition systems shape who gets to contribute and who does not. This supports the societal framing around inequality and cumulative advantage, and raises the question of whether best-paper awards similarly consolidate advantage among already well-connected junior researchers or open doors for those at the margins of elite networks.

2 Cumulative Advantage & the Matthew Effect

The Matthew Effect In Science

Merton coins the term "Matthew effect" to describe the disproportionate credit given to already eminent scientists compared to lesser-known researchers making equally significant contributions. Drawing on interviews with Nobel laureates and their collaborators, he argues that initial differences in scientific reputation compound over time, leading to a concentration of recognition and resources among a small elite [6].

This is the foundational theoretical paper for the entire project. The question at the heart of my thesis, whether conference awards primarily select researchers already on strong trajectories or actively amplify advantage is a direct operationalisation of Merton's original insight applied to a specific and understudied recognition mechanism conference best-paper awards in computer science.

Quantitative And Empirical Demonstration Of The Matthew Effect In A Study Of Career Longevity

Petersen et al. develop a stochastic model of career progress that formally incorporates the Matthew effect, testing it on publication records of 400,000 scientists and career records of over 20,000 athletes. They show that early career advantages compound over time through a "rich get richer" mechanism, and that many careers are stunted early by a relative disadvantage that proves difficult to overcome [7].

This paper provides a quantitative foundation for cumulative advantage theory directly applicable to my thesis. The finding that early career stage is decisive and that early disadvantages are hard to reverse, supports the design choice of focusing on junior researchers (within five years of first publication), as this is precisely the window where award recognition may have the longest-lasting effect.

The Matthew Effect In Science Funding

Using a natural experiment in Dutch research funding, where grant scores near a funding threshold are essentially random, Bol et al. demonstrate that researchers who narrowly receive a prestigious grant subsequently attract significantly more funding than equally qualified peers who narrowly missed it. This is among the cleanest causal demonstrations of the Matthew effect in science [8].

The regression-discontinuity design used here is methodologically instructive for my thesis. While my study uses matching rather than a discontinuity, the underlying logic isolating the effect of recognition from pre-existing quality is the same. The finding that even marginal funding differences create large long-term divergences strengthens the motivation for studying whether conference awards, even lower-stakes than major grants, operate similarly.

3 Early-Career Predictors of Long-Term Impact

Early-Career Factors Largely Determine The Future Impact Of Prominent Researchers: Evidence Across Eight Scientific Fields

Krauss et al. demonstrate, using longitudinal bibliometric data across eight fields, that a small number of early-career features notably early citation counts, institutional prestige at career start, and collaborations with already prominent researchers strongly predict long-term scientific impact decades later. The study finds that these early indicators are more predictive than later-career features, suggesting a durable "early advantage" mechanism [9].

This paper is critical to the thesis's framing of the selection versus amplification question. If best-paper awards are strongly correlated with the same early indicators that already predict success, this suggests awards primarily select high-potential researchers rather than creating new advantages. Controlling for early-career features in the regression analyses is directly motivated by these findings.

Quantifying The Evolution Of Individual Scientific Impact

Sinatra et al. analyse the publication histories of thousands of scientists across multiple disciplines and find a "random-impact rule", the highest-impact work in a scientist's career is equally likely to occur at any career stage. They develop a Q-model that decomposes impact into a stable individual parameter Q (intrinsic ability), productivity, and luck, finding that high-impact work requires both high Q and a lucky draw, not merely sustained productivity [10].

The random-impact rule directly challenges a naive interpretation of career trajectory analysis that impact simply rises with seniority. For my thesis, it raises the question of whether award-winning junior authors have demonstrably higher Q than matched controls, or whether their award-winning paper is itself partly a lucky draw. It also motivates longitudinal rather than cross-sectional analysis of post-award output.

The Misleading Narrative Of The Canonical Faculty Productivity Trajectory

Way et al. analyse the publication and employment histories of 2,453 tenure-track computer science faculty across all PhD-granting departments in the US and Canada. They show that the conventional narrative of "rapid rise followed by gradual decline" in productivity describes only about 20% of faculty, with the remaining 80% exhibiting highly diverse individual trajectories. They also find that departmental prestige predicts overall productivity and the timing of the transition from first- to last-author publications [11].

This paper is directly applicable to my thesis in two ways. First, the dataset is specifically from computer science, the same field studied here. Second, the finding that the first-to-last-author transition timing is predicted by departmental prestige is relevant to interpreting junior awardee author-position patterns. The diversity of individual trajectories also cautions against relying on group averages alone when comparing awardees to controls.

4 Team Size, Disruption & Collaboration

Large Teams Develop And Small Teams Disrupt Science And Technology

Wu et al. analyse over 65 million papers and patents and show that small research teams are systematically more likely to produce disruptive contributions, those that redirect scientific attention away from prior work while large teams are more likely to consolidate and develop existing research streams. They introduce the CD index as a measure of this distinction, and find that the relationship holds across fields, time periods, and types of knowledge output [12].

This paper introduces the core measurement tool the CD index used in my thesis and provides the theoretical justification for examining disruption as a dimension of scientific impact distinct from citation counts. The finding that small teams produce more disruptive work is directly relevant because best-paper award-winning teams in CS are often small, raising the question of whether junior awardees continue to produce disruptive work or shift toward larger, more consolidating teams post-award.

Papers And Patents Are Becoming Less Disruptive Over Time

Park et al. apply the CD index to 45 million papers and 3.9 million patents spanning several decades and find a robust, universal decline in the disruptiveness of scientific and technological output over time. They attribute this trend in part to researchers drawing on an increasingly narrow slice of existing literature, contradicting the expectation that a growing knowledge base would produce more diverse and disruptive ideas [13].

This paper establishes an important temporal baseline for disruption measurement in science. For my thesis, it implies that any comparison of awardees and non-awardees using the CD index should control for publication year, since disruption scores are systematically declining across the board. It also raises the question of whether best-paper award winners, who presumably produce high-impact work are more resistant to this secular decline in disruption.

Dimensions: Calculating Disruption Indices At Scale

Sixt and Pasin present a computational pipeline for calculating the CD index and related disruption metrics at large scale using the Dimensions bibliometric database. They discuss practical implementation challenges including citation completeness, reference list accuracy, and computational efficiency, and validate their approach against prior published results [14].

This paper is directly useful for the implementation phase of my thesis. When computing CD indices for the matched awardee and non-awardee cohorts using OpenAlex data, the same computational considerations apply. The authors' discussion of data quality issues such as incomplete reference lists affecting disruption scores is directly relevant to methodological decisions about data filtering and robustness checks.

The Increasing Dominance Of Teams In Production Of Knowledge

Wuchty et al. use 19.9 million papers and 2.1 million patents to show that teams have increasingly dominated the production of scientific knowledge over the past five decades, replacing individual scientists as the primary producers of high-impact research. They find that team-produced papers receive more citations on average, and that this citation advantage of teams has grown over time [15].

This paper provides essential context for my thesis's analysis of co-authorship patterns. Since the study period (2000-2018) spans a period of growing team science, changes in collaboration patterns for awardees versus controls must be interpreted against this broad trend. The finding that teams now produce exceptional impact work, once the domain of solo authors directly motivates examining whether junior awardees use their recognition to grow collaborative networks.

5 Bibliometric Methods & Data

Software Survey: VOSviewer, A Computer Program For Bibliometric Mapping

Van Eck and Waltman introduce VOSviewer, a freely available software tool for constructing and visualising bibliometric maps. The paper describes the program's core functionality, including support for co-authorship, co-citation, bibliographic coupling, and keyword co-occurrence analyses at multiple levels of aggregation such as author, journal, and country. The authors demonstrate the tool's scalability by constructing a co-citation map of 5,000 major scientific journals, arguing that VOSviewer's particular strength lies in the quality of its graphical output and its ability to handle large-scale datasets [16].

For my thesis, VOSviewer is a candidate tool for visualising co-authorship network evolution and citation-based clustering of awardees and non-awardees across conferences and time periods. Even if the primary analysis environment is Python and NetworkX, understanding VOSviewer's analytical categories particularly how it operationalises bibliographic coupling and co-authorship links helps ground the network metrics computed in this study. The paper also provides a methodological benchmark for how collaboration structure has been visualised in prior science-of-science work.

How To Conduct A Bibliometric Analysis: An Overview And Guidelines

Donthu et al. provide a systematic, step-by-step guide to conducting bibliometric analysis, covering both performance analysis and science mapping. The paper distinguishes bibliometric analysis from meta-analysis and systematic literature reviews, arguing it is best suited for large, broad corpora where manual review is infeasible. The authors outline key techniques citation analysis, co-citation analysis, bibliographic coupling, and co-word analysis and offer decision guidelines for selecting the appropriate method based on the research question and dataset characteristics [17].

This paper is relevant to the thesis as a methodological reference that justifies the choice of bibliometric indicators publication counts, citation rates, and co-authorship measures used to compare awardee and non-awardee trajectories. Its framing of bibliometrics as a tool for mapping intellectual structure and identifying trends directly applies to the thesis's aim of reconstructing longitudinal career patterns from OpenAlex metadata. The guidelines also help structure the analytical pipeline, particularly for decisions around which citation-based metrics to prioritise at different stages of the analysis.

Bibliometric Methods In Management And Organization

Zupic and Čater offer a comprehensive review and workflow guide for applying bibliometric methods, introducing five core techniques: citation analysis, co-citation analysis, bibliographic coupling, co-author analysis, and co-word analysis. The paper synthesises guidelines from 81 bibliometric studies and argues that these methods introduce systematic, transparent, and reproducible review processes that complement qualitative approaches. The authors present a recommended workflow covering database selection, data cleaning, analysis, and interpretation, illustrated with a full worked example [18].

For my thesis, Zupic and Čater's workflow is particularly useful as a reference standard for the data pipeline, from retrieving OpenAlex records through to computing and interpreting co-authorship and citation metrics. Their treatment of co-author analysis and bibliographic coupling maps directly onto the thesis's network science component, where co-authorship diversity and brokerage roles are tracked over time for awardees and controls. The paper's emphasis on transparency and reproducibility also aligns with the thesis's open-science commitments, providing methodological precedent for documenting analytical choices clearly.

An Index To Quantify An Individual's Scientific Research Output

Hirsch proposes the h-index, defined as the largest number h such that a researcher has h papers each receiving at least h citations, as a single-number summary of cumulative research output. The paper argues the h-index is more robust than total publication count, total citations, or citations per paper alone, because it simultaneously captures both productivity and sustained impact. Hirsch also discusses the index's limitations, including its sensitivity to field-specific citation norms and its tendency to grow monotonically with career length, making cross-field comparisons unreliable without normalisation [19].

In my thesis, the h-index may serve as one of the field-normalised impact indicators used to characterise career trajectories of junior awardees and non-awardees before and after the award event, and its original formulation is the essential citation for any study that uses or critiques it. Crucially, the paper's own caveats about field-specificity and career-stage dependence directly motivate the thesis's use of additional complementary indicators such as annual citation rates and the CD disruption index. They also reinforce the decision to conduct stratified analyses by venue and subfield rather than relying on any single aggregate measure.

Comparing Bibliometric Statistics Obtained From The Web Of Science And Scopus

Archambault et al. conduct a large-scale comparison of macro-level bibliometric indicators derived from the Web of Science and Scopus, analysing over 14 million papers and 100 million citations across both databases. The study finds extremely high correlations ($R^2 >$ .99) between the two sources in terms of publication counts, citation counts, and country-level rankings, even when broken down by field. The authors conclude that bibliometric indicators of scientific production and impact are largely stable and database-independent at the aggregate level [20].

For my thesis, which relies exclusively on OpenAlex as its data source, this paper provides important methodological context for understanding the implications of database choice. While OpenAlex is not directly compared with WoS or Scopus here, the finding that aggregate bibliometric patterns are consistent across major databases supports the assumption that OpenAlex-derived metrics publication counts, citation trajectories, and co-authorship records are sufficiently representative for the comparative career analysis conducted here. It also prompts awareness of coverage gaps that OpenAlex may have for specific venues or time periods, which the thesis addresses by documenting disambiguation limitations and restricting analysis to well-covered computer science conferences.

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